Santhosh BV
Electrical Engineering
Indian Institute of Technology Bombay

24B1290
B.Tech.
Gender: Male
DOB: 26/12/2006

Examination	University	Institute	Year	CPI / % Credits
Graduation	IIT Bombay	IIT Bombay	2028	9.88 **Core:** 136 **| Total:** 136
Intermediate	CBSE	Sri Chaitanya Techno School	2024	98.20%
Matriculation	CBSE	Sharanalaya International School	2022	99.00%

Pursuing a Minor Degree in **Machine Learning & Data Science** from C-MInDS IITB

Scholastic Achievements

- Recognized with the **Institute Academic Prize** for academic excellence in the first year among **1300+** students *('25)*
- Secured the **Academic Proficiency** grade in **Microprocessors**, the sole awardee in a class of **120+** students *('26)*
- Secured **Academic Proficiency** grade in Programming for **Data Science** & **Economics** among **200+** students *('25)*
- Bagged a perfect **Semester Performance Index** (SPI) of **10** in the second semester at IIT Bombay *('25)*
- Received **Academic Proficiency** grade in **Linear Algebra & Differential Equations** among **330+** students *('25)*

Key Technical Projects

8051-Based IMD Frequency Analyzer
Apr '26 - May '26

Course Project: Microprocessors | Prof. Shalabh Gupta, Department of Electrical Engineering

- Optimized memory usage for selective spectral analysis on **8051** by replacing the DFT with the **Goertzel algorithm**
- Developed synchronized ADC/DAC operation using **single timer** to reduce jitter, **phase-accumulator** LUT-based DDS, quantization-aware frequency selection, and an integer square-root algorithm for IMD amplitude computation

CPU Design in Verilog
Oct '25 - Nov '25

Course Project: Digital Systems | Prof. Sachin B Patkar, Department of Electrical Engineering

- Designed and implemented an **8-bit single-cycle** Harvard CPU in Verilog supporting **23** instructions across various categories, including data movement, memory access, arithmetic, logical, relational, and complex control flow operations
- Extended the architecture to a **16-bit multi-cycle** Von Neumann CPU, redesigning the datapath to integrate FSM-based control logic and validated the design by executing **Bubble Sort**, confirming correct instruction sequencing

Computer Architecture
May '26 - Present

Summer of Science | Maths and Physics Club, IIT Bombay

- Analyzing modern processor architectures, combining out-of-order superscalar execution using **Tomasulo's algorithm**, reservation stations and reorder buffers with **cache hierarchy**, memory-system optimization and branch prediction
- Exploring GPU architectures, SIMD, vector processors to understand **data-level parallelism** for parallel computing

Zig-Zag Run Length Encoding
Aug '25 - Nov '25

Digital Circuits Laboratory | Prof. Saurabh Lodha, Department of Electrical Engineering

- Developed modular **VHDL** logic to perform Zig-Zag Run-Length Encoding on **8*8** matrices of **8-bit** values, mimicking the **JPEG compression** standard to effectively reduce data redundancy through serialized coefficient scanning
- Utilized a **four-state FSM** to coordinate compression and decompression cycles, successfully validating the hardware architecture through **FPGA** implementation and comprehensive functional logic simulation using ModelSim Altera

I^2C-Based Multi-Sensor Interface
Apr '26

Microprocessors Laboratory | Prof. Sachin B Patkar, Department of Electrical Engineering

- Developed a reusable **I^2C driver** on the 8051 based AT89C5131A to interface BMP280- temperature sensor, and DS1307 RTC modules, enabling synchronized communication with multiple peripherals over a shared **two-wire bus**
- Implemented real-time processing of 20-bit calibrated sensor data with concurrent LCD and **UART** communication

Autonomous Mobile Manipulator
Dec '25 - Jan '26

winteROS | Electronics and Robotics Club, IIT Bombay

- Developed a unified **exploratory robot** in Gazebo featuring an autonomous navigation stack driven by **SLAM and Nav2**, utilizing fused **LiDAR**, **camera** sensor streams for obstacle avoidance, localization and dynamic path planning
- Integrated a **4-axis robotic arm** with dual-prismatic gripper and inverse kinematics, OpenCV-based object detection

RC Quadcopter
Mar '25 - Apr '25

Course Project: MS101 | Prof. Joseph John, Department of Electrical Engineering

- Collaborated in a team of **four** to build a quadcopter and achieved flight time of **30+ s** with controlled maneuverability
- Designed and analysed all the mechanical components using **Fusion360**, 3D printing the quadcopter parts in **PLA**
- Designed and fabricated a **custom PCB** integrating joystick and peripheral controls using **EasyEDA** to establish seamless wireless communication between the quadcopter and remote via 2.4GHz Wi-Fi using dual **ESP32** modules

Analog Differential Equation Solver
Apr '26

Analog Laboratory | Prof. Sandip Mondal, Department of Electrical Engineering

- Designed and implemented an analog differential equation solver to model **second-order free-fall dynamics**, validating conditions of free fall, terminal velocity, and oscillatory motion through simulation and hardware testing
- Utilized **Howland current-source** integrator to enable configurable initial conditions and electrolytic capacitor usage

Other Projects

Embedded Sound Experimentation System
Jan '26 - Apr '26

Course Project: Design Thinking for Innovation | Prof. Nishant Sharma, IDC

- Developed in a team of 6 a **sound experiment module**, applying the Double Diamond **design thinking** process
- Integrated a Raspberry Pi Pico, INMP441 microphone, servos, and a Streamlit web interface to build an **embedded system** for sound experiments,demonstrating frequency, resonance, and timbre through real-time sensing, actuation

XLR8 : Bot Racing Competition
Aug '24 - Sep '24

Electronics and Robotics Club, IIT Bombay

- Tinkered, designed, and thoroughly tested in a team of four a Wi-Fi controlled **RC car** with an acrylic chassis, 200 RPM BO DC motors, and **L298N** driver powered by **Raspberry Pi Pico W** for versatile multi-terrain maneuvering
- Integrated **MPU6050** with an **ESP32**-based remote to capture orientation inputs and wirelessly transmit commands to Pico W, where wheel speed control algorithms were programmed and optimized to achieve precise motion control

Object Localisation using Wildlife Images
Oct '25 - Nov '25

Course Project: Programming for Data Science | Prof. Vinay Kulkarni, C-MInDS

- Achieved a **0.85 Test F1-score** using an **XGBoost** classifier for wildlife localization, leveraging **47 handcrafted** features including Gabor, Sobel filters, saliency maps, HSV analysis, **block coordinates** to capture spatial context
- Localized wildlife on 8×8 grid using cell, neighborhood spatial descriptors, trained on semi-automatically labeled data

Strategic Planning & Analysis
Sep '24 - Nov '24

Course Project: Introduction to Management | Prof. Mayank Pareek, Shailesh J. Mehta School of Management

- Collaborated with a **team of ten** to develop strategic plans for Nestle, analyzing mission, vision, values, and conducting a SWOT analysis to create effective strategies for long-, short-term, **value chain** optimization and **budget** allocation
- Analyzed industry competition using **Porter's Five Forces**, **VRIO** framework to assess leverage of Nestle's products

Financial Derivatives & Time Series Analysis
May '26 - Jul '26

Reading Project: Summer of Quant |Quant Club, IIT Bombay

- Developed a comprehensive understanding of **financial derivatives**, their **pricing** using no-arbitrage, Black–Scholes, binomial trees, cost-of-carry models, and their application in hedging and risk management through futures and swaps
- Analyzed time series using **ARIMA**, GARCH models and employed diagnostic tests for volatility modeling, forecasting

Technical Skills

Programming	C++, Python, Verilog, VHDL, Embedded-C, 8051 Assembly
Softwares and Packages	Quartus, LTspice, EasyEDA, Keil µVision, Fusion360, Fracktory, LaTeX
Python Libraries	NumPy, Pandas, Matplotlib, SciPy, Seaborn, Sklearn, Statsmodels, pyTorch, OpenCV

Extracurriculars

- Attended a 5-day bootcamp on **communication**, **teamwork**, and leadership conducted by Career Cell, IITB ('25)
- Built an **RC plane** using Depron, powered by BLDC motors with ESC and servo controls for pitch and roll. Competed and recorded fly time of **10+s** in the RC Plane Competition organized by the Aeromodelling Club, IIT Bombay ('24)
- Participated in building an autonomous **Mini Militia bot** for Code Wars organized by WNCC, IIT Bombay ('26)
- Selected from over **1000+** students, in the first year, for year-long **basketball** training under **NSO**, IIT Bombay ('25)
- Participated in the department basketball competition organized by Electrical Engineering Students' Association ('24)
- Designed a multipurpose pen stand, optimized **hostel room organization** in the Introduction to Design course ('24)